



TARGET JUPITER

The Galileo space probe was launched by the Shuttle Atlantis in October 1989. After a six-year journey, it approached Jupiter in July 1995. While the main probe went into orbit around the planet, a smaller atmospheric probe descended into Jupiter's atmosphere. It relayed data back to Earth for 57 minutes before being destroyed by the pressure of Jupiter's clouds. The quality of the information provided by Galileo and other unmanned space probes has revolutionized astronomers' understanding of the planets.

The HST went on to prove itself one of the greatest science success stories of the modern era, supplying astonishing images of distant galaxies. There could no longer be any question of the Shuttle's importance in launching and servicing satellites. But the very success of such unmanned scientific missions as the *Galileo* probe and the HST called into question the need for a manned space program, except as a subsidiary service to maintain unmanned satellites.

Continuing goal

Yet the goal of manned space flight continued to be pursued, if fitfully as arguments against its high cost intermittently blocked the way forward. In favor of manned flight it was urged that scientists in space could perform experiments that could not be done remotely. It was said that putting human beings in space was in itself a valuable experiment – when 77-year-old John Glenn traveled in the Shuttle in 1998 this was presented as a test of the effects of space travel on an older body, that might ultimately benefit senior citizens on Earth. And beyond all practical arguments was the sense of a final frontier that must always be pushed further back, the

exploratory imperative – if humans were capable of visiting Mars, how could they resist doing it?

The early 1980s brought a revival of tension between the superpowers that favored prestigious space projects as it had in the 1960s.

President Ronald Reagan not only committed himself to the

The report of the Rogers Commission, set up to inquire into the *Challenger* accident, concluded that NASA had been under such pressure to maintain the schedule of Shuttle launches that the agency had ceased to give safety and quality control the priority they required. While reaffirming the importance of NASA and the space program as “symbols of national pride and technological leadership,” the commission called for a thorough safety review before Shuttle launches resumed at a more manageable rhythm.

Return of the Shuttle

The Shuttle returned to space in September 1988 and the program worked up to a launch every six to eight weeks. But the vision of the Shuttle as a self-financing part of the commercial space business was at an end. It would now carry only loads specifically designed for it or essential military space hardware.

Nor was the Shuttle any longer expected to replace disposable launch systems for satellites. The traditional rocket launchers had themselves gone through a difficult period – in 1986 a USAF Titan rocket blew up eight seconds after launch and a NASA Delta launcher also failed early in flight. But the rockets were there to stay, working alongside the reusable space vehicle.

Projects delayed by the interruption of the Shuttle programme included the *Galileo* space

probe designed to fly to Jupiter. It had been scheduled for launch in 1986 and had actually been transported to Cape Canaveral when the *Challenger* accident occurred. *Galileo* returned there in 1989 and was taken into space by the Shuttle *Atlantis*.

Also waiting for Shuttle launch in 1986 had been the Hubble Space Telescope (HST). Taking its place in line, the HST went into orbit in April 1990. At first, the mission appeared to be another expensive embarrassment for NASA. The great mirror designed to gather light from near to the origin of the universe did not work well because of a minor grinding error. Out of adversity, however, came a resounding triumph. In December 1993, the Shuttle *Endeavour* met up with the HST in orbit and grabbed it with a manipulator arm. Members of the Shuttle crew then exited the vehicle to carry out a grueling repair job, their space-suited figures shown to the world in another space spectacular.



VETERAN RETURNS

The return of veteran astronaut John Glenn to space in 1998 at the age of 77 was one of NASA's most successful publicity coups. Glenn had completed his first space mission in 1962 aboard *Friendship 7*.